

# **Chapter 9: Inferences Based on Two Samples**

Probability and Statistics for Science and Engineering  
with Examples in R

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# (1) Inferences Concerning Two means

Setting:

1.  $X_1, \dots, X_m$  is a random sample of size  $m$  from population 1 which has mean  $\mu_1$  and variance  $\sigma_1^2$ .
2.  $Y_1, \dots, Y_n$  is a random sample of size  $n$  from population 2 which has mean  $\mu_2$  and variance  $\sigma_2^2$ .
3. The two samples  $X_1, \dots, X_m$  and  $Y_1, \dots, Y_n$  are independent.

Inferences will be made about the difference in means:

$$\mu_1 - \mu_2$$

# (A) Large, Independent Samples

(Both  $m, n \geq 30$ )

$$\left. \begin{array}{l} 1. E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = \Delta \\ 2. Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n} \end{array} \right\} Z = \frac{(\bar{X} - \bar{Y}) - \Delta}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}} \rightarrow N(0,1)$$

 Independence  CLT

When the sample sizes  $m$  and  $n$  are large ( $m, n \geq 30$ ), the normal approximation remains valid when sample variances replace the population variances. Consequently, the two sample  $Z$  statistic becomes

$$Z = \frac{(\bar{X} - \bar{Y}) - \Delta}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \rightarrow N(0,1).$$

## (a) CI

$$P\left(-z_{\alpha/2} < \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}} < z_{\alpha/2}\right) \approx 1 - \alpha$$

$$P\left(-z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} < (\bar{X} - \bar{Y}) - (\mu_1 - \mu_2) < z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}\right) \approx 1 - \alpha$$

$$P\left((\bar{X} - \bar{Y}) - z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} < \mu_1 - \mu_2 < (\bar{X} - \bar{Y}) + z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}\right) \approx 1 - \alpha$$

For two independent populations, a  $100(1 - \alpha)\%$  CI for  $\mu_1 - \mu_2$ :

$$\left( (\bar{x} - \bar{y}) - z_{\frac{\alpha}{2}} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}, (\bar{x} - \bar{y}) + z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} \right)$$

If  $m \geq 30$  and  $n \geq 30$

Example 9.1: In June 2024, chemical analyses were made of 85 water samples (each of unit volume) taken from various parts of a city lake, and the measurements of chlorine content were recorded. During the next two winters, the use of road salt was substantially reduced in the catchment areas of the lake. In June 2026, 110 water samples were analyzed.

|                    | Chlorine Content (in ppm) |      |
|--------------------|---------------------------|------|
|                    | 2024                      | 2026 |
| Mean               | 18.3                      | 17.8 |
| Standard Deviation | 1.2                       | 1.8  |

Construct a 95% CI for the difference of the population means.

Since both  $m = 85$  and  $n = 110$  are large, we can use the following formula.

$$\begin{aligned}
 \bar{x} - \bar{y} \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} &= 18.3 - 17.8 \pm z_{0.025} \sqrt{\frac{(1.2)^2}{85} + \frac{(1.8)^2}{110}} \\
 &= 0.5 \pm 1.96(0.2154) = 0.5 \pm 0.422
 \end{aligned}$$

Thus, a 95% confidence interval for  $\mu_1 - \mu_2$  is (0.078, 0.922).

0 is not included  $\rightarrow$  significant difference

# Testing hypothesis

|               | Case (a)                                                                                                                                  | Case (b)                                                                                                                              | Case (c)                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1</b> | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 \neq \Delta_0$<br>(2-sided alternative)                                           | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 > \Delta_0$<br>(1-sided alternative)                                          | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 < \Delta_0$<br>(1-sided alternative)                                           |
| <b>Step 2</b> | $\alpha = ?$                                                                                                                              | $\alpha = ?$                                                                                                                          | $\alpha = ?$                                                                                                                           |
| <b>Step 3</b> | $z = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <u>Rejection Region:</u><br>$ z  \geq z_{\alpha/2}$ | $z = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <u>Rejection Region:</u><br>$z \geq z_{\alpha}$ | $z = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <u>Rejection Region:</u><br>$z \leq -z_{\alpha}$ |
| <b>Step 4</b> | Substitute $\bar{x}, \bar{y},$<br>$s_1^2, s_2^2, m$ and $n$<br>Calculate $z$<br>Decision                                                  | Substitute $\bar{x}, \bar{y}, s_1^2, s_2^2, m$<br>and $n$<br>Calculate $z$<br>Decision                                                | Substitute $\bar{x}, \bar{y},$<br>$s_1^2, s_2^2, m$ and $n$<br>Calculate $z$<br>Decision                                               |
| <b>Step 5</b> | $p = 2P(Z \geq  z )$                                                                                                                      | $p = P(Z \geq z)$                                                                                                                     | $p = P(Z \leq z)$                                                                                                                      |

Example 9.2: From Example 9.1, do the data provide strong evidence that there is a reduction of average chlorine level in the lake water in 2026 compared to the level in 2024? Test with  $\alpha = 0.05$ .

$$H_0: \mu_1 = \mu_2, H_1: \mu_1 > \mu_2$$
$$\alpha = 0.05$$

$$\text{Test statistic: } Z = \frac{\bar{X} - \bar{Y} - 0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$$

$$\text{Rejection region: } z \geq z_\alpha = z_{0.05} = 1.645$$

$$z = \frac{18.3 - 17.8}{\sqrt{\frac{(1.2)^2}{85} + \frac{(1.8)^2}{110}}} = 2.32 > 1.645$$

Reject  $H_0$

→ Yes, the data provide strong evidence that there is a reduction of average chlorine level in the lake water in 2026 compared to 2024.

$$p\text{-value} = P(Z > 2.32) = 1 - \Phi(2.32) = 0.01$$

Using R to find P-value,  
>1-pnorm(2.32)

# **(B) Small, Independent Samples**

$(m < 30 \text{ and/or } n < 30)$

Assumption: Both populations are normal

## (a) When we assume $\sigma_1 = \sigma_2 = \sigma$

$$\left. \begin{array}{l} 1. E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = \Delta \\ 2. Var(\bar{X} - \bar{Y}) \stackrel{\text{Independence}}{=} Var(\bar{X}) + Var(\bar{Y}) = \sigma^2 \left( \frac{1}{m} + \frac{1}{n} \right) \end{array} \right\} Z = \frac{(\bar{X} - \bar{Y}) - \Delta}{\sigma \sqrt{\left( \frac{1}{m} + \frac{1}{n} \right)}} \sim N(0,1)$$

Independence

The unknown  $\sigma^2$  is estimated by the following pooled estimator:

$$S_p^2 = \frac{(m-1)S_1^2 + (n-1)S_2^2}{m+n-2}$$

Test statistic:

$$T = \frac{(\bar{X} - \bar{Y}) - \Delta}{S_p \sqrt{\left( \frac{1}{m} + \frac{1}{n} \right)}} \sim t_{m+n-2}$$

$$\text{where } S_p^2 = \frac{(m-1)S_1^2 + (n-1)S_2^2}{m+n-2}$$

For two independent populations, a  $100(1 - \alpha)\%$  CI for  $\mu_1 - \mu_2$ :

$$\left( \bar{x} - \bar{y} - t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}}, \bar{x} - \bar{y} + t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}} \right)$$

If  $m < 30$  and/or  $n < 30$

Example 9.3: A random sample of 15 mothers with low socioeconomic status deliver babies whose average birthweight was 110 ounces and a sample standard deviation was 24 ounces, whereas a random sample of 76 mothers with a medium socioeconomic status resulted in a sample average birthweight and sample standard deviation of 120 ounces and 22 ounces, respectively. Assume that the average birthweights are normally distributed in both populations. Find a 95% CI of  $\mu_1 - \mu_2$ .

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{14(24)^2 + 75(22)^2}{15+76-2} = 498.5$$

$$t_{\frac{\alpha}{2}, m+n-2} = t_{0.025, 89} = 1.99$$

$$\bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}} = 110 - 120 \pm 1.99 \sqrt{498.5} \sqrt{\frac{1}{15} + \frac{1}{76}} = -10 \pm 12.5$$

95% confidence interval for  $\mu_1 - \mu_2$ :  $(-22.5, 2.5)$

# Testing hypothesis

|        | Case (a)                                                                                                                                             | Case (b)                                                                                                                                 | Case (c)                                                                                                                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1 | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 \neq \Delta_0$<br>(2-sided alternative)                                                      | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 > \Delta_0$<br>(1-sided alternative)                                             | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 < \Delta_0$<br>(1-sided alternative)                                              |
| Step 2 | $\alpha = ?$                                                                                                                                         | $\alpha = ?$                                                                                                                             | $\alpha = ?$                                                                                                                              |
| Step 3 | $t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$ <u>Rejection Region:</u><br>$ t  \geq t_{\frac{\alpha}{2}, m+n-2}$ | $t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$ <u>Rejection Region:</u><br>$t \geq t_{\alpha, m+n-2}$ | $t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$ <u>Rejection Region:</u><br>$t \leq -t_{\alpha, m+n-2}$ |
| Step 4 | Substitute $\bar{x}, \bar{y}, s_p, m$<br>and $n$<br>Calculate $t$<br>Decision                                                                        | Substitute $\bar{x}, \bar{y}, s_p, m$ and<br>$n$<br>Calculate $t$<br>Decision                                                            | Substitute $\bar{x}, \bar{y}, s_p, m$<br>and $n$<br>Calculate $t$<br>Decision                                                             |
| Step 5 | $p = 2P(T \geq  t )$                                                                                                                                 | $p = P(T \geq t)$                                                                                                                        | $p = P(T \leq t)$                                                                                                                         |

Example 9.4: IQ's of 16 students from one area of a city:  $\bar{x} = 107, s_1 = 10$ ; IQ's of 14 students from another area of the city:  $\bar{y} = 112, s_2 = 8$ . The IQ scores are assumed to be normally distributed. Is there a significant difference between the IQ's of the two groups at significance level of 0.01?

$$H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2$$

$$\alpha = 0.01$$

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{15(10)^2 + 13(8)^2}{16+14-2} = 83.3$$

$$\text{Test statistic: } T = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$$

$$\text{Rejection region: } |t| \geq t_{\frac{\alpha}{2}, m+n-2} = t_{0.005, 28} = 2.763$$

$$t = \frac{107 - 112}{\sqrt{83.3} \sqrt{\frac{1}{16} + \frac{1}{14}}} = -1.496$$

Since  $|t| = 1.496 < 2.763$ , we do not reject  $H_0$ . There is not a significant difference between the IQ's of the two groups at level 0.01.

$$t_{0.1, 28} = 1.313 < 1.496 < 1.701 = t_{0.05, 28}$$

$$0.1 = 2 \times 0.05 < p\text{-value} < 2 \times 0.1 = 0.2$$

## Example 9.4 (continued)

To calculate the p-value using R,

```
>2*(1-pt(1.496, 28))
```

```
0.1458411
```

To conduct hypothesis testing using R,

```
>t.test(x, y, conf.level=0.01,var.equal=T)
```

The output contains a 99% confidence interval for the mean difference, the value of the t statistic and the *p*-value.

## (b) When we assume $\sigma_1 \neq \sigma_2$

$$\text{CI: } \bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}, \nu} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}$$

$$\text{where } \nu = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}} \text{ (round down)}$$

$$\text{Test statistic: } T = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \sim t \text{ with df} = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$$

For two independent normal populations with  $\sigma_1 \neq \sigma_2$ , a  $100(1 - \alpha)\%$  CI for  $\mu_1 - \mu_2$ :

$$\left( \bar{x} - \bar{y} - t_{\alpha/2, v} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}, \bar{x} - \bar{y} + t_{\alpha/2, v} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} \right), \text{ where } v = \frac{\left( \frac{s_1^2}{m} + \frac{s_2^2}{n} \right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$$

Example 9.5: Breaking load (pound/inch width) for two kinds of fabrics

| Fabric type | Sample Size | Sample Mean | Sample sd |
|-------------|-------------|-------------|-----------|
| Cotton      | 10          | 25.9        | 0.4       |
| Triacetate  | 10          | 68.1        | 1.8       |

Assuming normality, find a 95% CI for the difference between true average porosity for the two types of fabric.

$$v = \frac{\left( \frac{s_1^2}{m} + \frac{s_2^2}{n} \right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}} = \frac{\left( \frac{(0.4)^2}{10} + \frac{(1.8)^2}{10} \right)^2}{\frac{((0.4)^2/10)^2}{9} + \frac{((1.8)^2/10)^2}{9}} = 9.89 \rightarrow v = 9$$

$$\bar{x} - \bar{y} \pm t_{0.025, 9} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} = 25.9 - 68.1 \pm 2.262 \sqrt{\frac{(0.4)^2}{10} + \frac{(1.8)^2}{10}} = -42.2 \pm 1.32$$

95% CI:  $(-43.52, -40.88)$

# Testing hypothesis

|        | Case (a)                                                                                                                                                                                                                                                           | Case (b)                                                                                                                                                                                                                                               | Case (c)                                                                                                                                                                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1 | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 \neq \Delta_0$<br>(2-sided alternative)                                                                                                                                                                    | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 > \Delta_0$<br>(1-sided alternative)                                                                                                                                                           | $H_0: \mu_1 - \mu_2 = \Delta_0,$<br>$H_1: \mu_1 - \mu_2 < \Delta_0$<br>(1-sided alternative)                                                                                                                                                            |
| Step 2 | $\alpha = ?$                                                                                                                                                                                                                                                       | $\alpha = ?$                                                                                                                                                                                                                                           | $\alpha = ?$                                                                                                                                                                                                                                            |
| Step 3 | $t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <p>Rejection Region:</p> $ t  \geq t_{\frac{\alpha}{2}, v}$ $v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$ | $t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <p>Rejection Region:</p> $t \geq t_{\alpha, v}$ $v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$ | $t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <p>Rejection Region:</p> $t \leq -t_{\alpha, v}$ $v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$ |
| Step 4 | Substitute $\bar{x}, \bar{y},$<br>$s_1^2, s_2^2, m$ and $n$<br>Calculate $t$<br>Decision                                                                                                                                                                           | Substitute $\bar{x}, \bar{y},$<br>$s_1^2, s_2^2, m$ and $n$<br>Calculate $t$<br>Decision                                                                                                                                                               | Substitute $\bar{x}, \bar{y},$<br>$s_1^2, s_2^2, m$ and $n$<br>Calculate $t$<br>Decision                                                                                                                                                                |
| Step 5 | $p = 2P(T \geq  t )$                                                                                                                                                                                                                                               | $p = P(T \geq t)$                                                                                                                                                                                                                                      | $p = P(T \leq t)$                                                                                                                                                                                                                                       |

Example 9.6: The lengths of possession (in months) by first owner were observed for mid-size sedan models from two auto companies. The company making Model B cars claims that the owners like its cars better than the cars of Model A.

|                |    |    |    |    |    |    |    |    |    |    |
|----------------|----|----|----|----|----|----|----|----|----|----|
| <b>Model A</b> | 55 | 54 | 53 | 56 | 50 | 63 | 65 | 62 | 64 | 55 |
| <b>Model B</b> | 60 | 67 | 67 | 65 | 62 | 58 | 58 | 58 |    |    |

Assume  $\sigma_1 \neq \sigma_2$  and carry out a test to see whether the data support this conclusion using  $\alpha = 0.05$ .

### Example 9.6 (continued)

$m = 10, n = 8$ : small,  $\bar{x} = 57.70, \bar{y} = 61.88, s_1 = 5.29, s_2 = 3.98$

$H_0: \mu_1 - \mu_2 = 0, H_1: \mu_1 - \mu_2 < 0$

$\alpha = 0.05$

$$v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}} = \frac{\left(\frac{(5.29)^2}{10} + \frac{(3.98)^2}{8}\right)^2}{\frac{((5.29)^2/10)^2}{9} + \frac{((3.98)^2/8)^2}{7}} = 15.97 \rightarrow v = 15$$

$$\text{Test statistic: } T = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$$

Rejection region:  $t < -t_{\alpha, v} = -t_{0.05, 15} = -1.753$

$$t = \frac{57.70 - 61.88}{\sqrt{\frac{(5.29)^2}{10} + \frac{(3.98)^2}{8}}} = -1.912 < -1.753 \rightarrow \text{Reject } H_0$$

→ There is a significant difference between the lengths of time that owners keep cars of Model B compared to Model A at level 0.05 (Model B cars are kept for longer) .

$$t_{0.05, 15} = 1.753 < 1.912 < 2.131 = t_{0.025, 15}$$

$$0.025 < p\text{-value} < 0.05$$

## Example 9.6 (continued)

To calculate the  $p$ -value using R,

```
>pt(-1.912, 15)
```

```
0.03758497
```

To conduct hypothesis testing using R,

```
>x=c(55, 54, 53, 56, 50, 63, 65, 62, 64, 55)
```

```
>y=c(60, 67, 67, 65, 62, 58, 58, 58)
```

```
>t.test(x, y, alt= "less",var.equal=F)
```

To construct a 95% CI in Example 9.5 using R,

```
>t.test(x, y, var.equal=F)
```

## (C) Paired Data

For paired data, a  $100(1 - \alpha)\%$  confidence interval for the mean difference  $\delta$ :

$$\left( \bar{d} - t_{\frac{\alpha}{2}, n-1} \frac{s_d}{\sqrt{n}}, \bar{d} + t_{\frac{\alpha}{2}, n-1} \frac{s_d}{\sqrt{n}} \right) \text{ for } n < 30$$

Example 9.7: Two different methods for determining chlorine content (mg/L) on samples of  $\text{Cl}_2$ -demand-free water for various doses & contact times

|           | Subject |       |       |       |       |       |       |       |
|-----------|---------|-------|-------|-------|-------|-------|-------|-------|
|           | 1       | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
| Method I  | 0.39    | 0.84  | 1.76  | 3.35  | 4.69  | 7.70  | 10.52 | 10.92 |
| Method II | 0.36    | 1.35  | 2.56  | 3.92  | 5.35  | 8.33  | 10.70 | 10.91 |
| $d$       | 0.03    | -0.51 | -0.80 | -0.57 | -0.66 | -0.63 | -0.18 | 0.01  |

$$\bar{d} = -0.414, s_d = 0.321$$

(a) A 99% CI for the difference in true average chlorine readings between 2 methods:

$$\bar{d} \pm t_{\frac{\alpha}{2}, n-1} \frac{s_d}{\sqrt{n}} = \bar{d} \pm t_{0.005, 7} \frac{s_d}{\sqrt{n}} = -0.414 \pm 3.499 \frac{0.321}{\sqrt{8}} = -0.414 \pm 0.397$$

99% CI:  $(-0.811, -0.017)$

(b) Is there a significant difference in the true average chlorine readings?

0 is not included in the CI  $\rightarrow$  There is a significant difference in true average.

# Hypothesis testing

|        | Case (a)                                                       | Case (b)                                                    | Case (c)                                                    |
|--------|----------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| Step 1 | $H_0: \delta = 0, H_1: \delta \neq 0$<br>(2-sided alternative) | $H_0: \delta = 0, H_1: \delta > 0$<br>(1-sided alternative) | $H_0: \delta = 0, H_1: \delta < 0$<br>(1-sided alternative) |
| Step 2 | $\alpha = ?$                                                   | $\alpha = ?$                                                | $\alpha = ?$                                                |

If  $n < 30$ , conduct the  $t$ -test with  $n - 1$  degrees of freedom

| Step 3 | $t = \frac{\bar{d}}{s_d/\sqrt{n}}$ <u>Rejection Region:</u><br>$ t  \geq t_{\alpha/2}$ | $t = \frac{\bar{d}}{s_d/\sqrt{n}}$ <u>Rejection Region:</u><br>$t \geq t_{\alpha}$ | $t = \frac{\bar{d}}{s_d/\sqrt{n}}$ <u>Rejection Region:</u><br>$t \leq -t_{\alpha}$ |
|--------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Step 4 | Substitute $\bar{d}$ , $s_d$ and $n$<br>Calculate $t$<br>Decision                      | Substitute $\bar{d}$ , $s_d$ and $n$<br>Calculate $t$<br>Decision                  | Substitute $\bar{d}$ , $s_d$ and $n$<br>Calculate $t$<br>Decision                   |
| Step 5 | $p = 2P(T \geq  t )$                                                                   | $p = P(T \geq t)$                                                                  | $p = P(T \leq t)$                                                                   |

Example 9.8: A test is designed to compare the wearing qualities of two brands of motorcycle tires. One tire of each brand is placed on each of 6 motorcycles that are driven for a specified mileage. The two tires are then exchanged (front and rear) and driven for an equal number of miles. Wear is measured (in thousandths of an inch). Test for difference of wearing using  $\alpha = 0.05$ .

|         | Motorcycle |    |    |     |    |     |
|---------|------------|----|----|-----|----|-----|
|         | 1          | 2  | 3  | 4   | 5  | 6   |
| Brand A | 98         | 61 | 38 | 117 | 88 | 109 |
| Brand B | 102        | 60 | 46 | 125 | 94 | 111 |
| $d$     | -4         | 1  | -8 | -8  | -6 | -2  |

$$n = 6, \bar{d} = -4.5, s_d = 3.56$$

$$H_0: \delta = 0, H_1: \delta \neq 0$$

$$\alpha = 0.05$$

$$\text{Test statistic: } T = \frac{\bar{D}}{s_d/\sqrt{n}}$$

$$\text{Rejection region: } |t| \geq t_{\frac{\alpha}{2}, n-1} = t_{0.025, 5} = 2.571$$

$$t = \frac{-4.5}{3.56/\sqrt{6}} = -3.096$$

Since  $|t| = 3.096 > 2.571$ , we reject  $H_0 \rightarrow$  There is a significant difference at level 0.05.

$$t_{0.025, 5} = 2.571 < 3.096 < 3.365 = t_{0.01, 5}$$

$$0.01 \times 2 = 0.02 < p\text{-value} < 0.05 = 0.025 \times 2$$

## Example 9.8 (continued)

To calculate the p-value using R,

```
>2*pt(-3.096, 5)
```

```
0.02697522
```

To conduct hypothesis testing using R,

```
>t.test(x-y)
```

Alternatively,

```
>t.test(x, y, paired=T)
```

